MILITARY SPECIFICATION
STAND, FILL-DRAIN, COOLANT SYSTEM
MT-4730( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V) (referred to herein as the stand).

1.2 Classification. The stand shall consist of the following items:

Nomenclature	Reference Para.
a. Structure	3.6.1
b. Coolant hydraulic system	3.6.2
c. Coolant drain and storage system	3.6.3
d. Pressurization system	3.6.4
e. Work surface	3.6.5
f. Support and positioning mechanism	3.6.6

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) is the applicable general specification for the stand described herein (see 6.6).

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SPECIFICATIONS

Military

MIL-T-81734(AS)	Test Set, Airborne Weapon Control System AN/AWM-23(V)

Hughes Aircraft Company

GS30620-781	General Specification for Identification and Marking of AN/AWM-23 Ground Support Equipment

STANDARDS

Military

MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)

Hughes Aircraft Company

HP8-5	Hughes Process, Identification of Detail Parts and Assemblies
HP8-15	Hughes Process, Identification by the Silk Screen Method

DRAWINGS

Hughes Aircraft Company

493715-1XX	Stand, Fill-Drain, Coolant System

PUBLICATIONS

Hughes Aircraft Company

AR31082-531	Acceptance Requirements, Stand, Fill-Drain, Coolant System

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precednece shall apply:

a. Contract - The contract shall have precedence over any other document.

b. This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).

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c. Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Not applicable. Good commercial practice shall apply.

3.1.1 Selection of materials, parts, and processes. Not applicable.

3.1.1.1 Approval of nonstandard materials, parts, and processes. Not applicable. Nonstandard parts, materials and processes selected shall not require Government approval (see 6.3.15).

3.1.2 Flammable materials. Applicable.

3.1.3 Cables and connectors. Not applicable.

3.1.4 Wire. Not applicable.

3.1.5 Rotary switches. Not applicable.

3.1.6 Finishes. Applicable.

3.1.7 Marking and identification. Applicable except as follows. Panel marking shall be in accordance with HP8-15, Type 1, Class 1, or equivalent, and other marking in accordance with HP8-5, Type I, Class 3, Form 1, Grade A or equivalent.

3.1.8 Soldering. Not applicable.

3.1.9 Welding

3.1.9.1 Structural. Applicable, or equivalent.

3.1.9.2 Welds, resistance, electrical interconnections. Not applicable.

3.1.10 Printed wiring. Not applicable.

3.1.11 Engineering changes to printed circuit boards. Not applicable.

3.1.12 Replaceability of electronic standard parts and materials. Not applicable.

3.2 Design and construction. Applicable.

3.2.1 Design features. Not applicable.

3.2.1.1 Method of operation. Applicable. The stand shall be manually operable.

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3.2.1.2 Commercial test equipment. Not applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Not applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Not applicable. Product identification shall be in accordance with Drawing 493715-1XX and GS30620-781.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Applicable.

3.2.1.9 Space and weight limitations. The maximum size of the stand shall be 62 inches high by 48.5 inches wide by 37 inches deep. The maximum dry weight shall be 360 pounds.

3.2.1.10 Connectors. Not applicable.

3.2.1.11 Secondary electrical power. Not applicable.

3.2.1.12 Interface. Applicable. The stand shall interface with facility coolant supply (see 3.2.2.4), facility nitrogen supply (see 3.2.2.6), and with the AN/AWG-9 weapon replaceable assemblies (WRAs) and AN/AWM-23 test panels listed in Table I. The stand shall interface with Cooler, Liquid, Electronic Equipment HD-908 ( )/AWM-23(V), and HD-957 ( )/AWM-23(V) via facility coolant lines.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Not applicable.

3.2.2.2 Clean air. Not applicable.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Applicable, except the nitrogen supplied by the facility will be at a pressure regulated to 90 $\pm$5 psig.

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TABLE I

EQUIPMENT MAINTAINED AT THE COOLANT SYSTEM
DRAIN-FILL STAND

Item	Nomenclature (See 6.4)
1	Oscillator, Radio Frequency O-1633( )/AWG-9
2	Transmitter, Radar T-1224 ( )/AWG-9
3	Power Supply PP-6770 ( )/AWG-9
4	Power Supply PP-6771 ( )/AWG-9
5	Receiver, Radar R-1785 ( )/AWG-9
6	Receiver, Infrared R-1786 ( )/AWG-9
7	Processor, Data CV-2947 ( )/AWG-9
8	Dummy Load, Electrical DA-630 ( )/AWM-23(V)
9	Dummy Load, Electrical DA-631 ( )/AWM-23(V)
10	Dummy Load, Electrical DA-632 ( )/AWM-23(V)
11	Transducer TR-283 ( )/AWM-23(V)

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3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions

3.2.3.1 Electrical. Not applicable.

3.2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Not applicable. MIL-STD-470 shall serve as a guide.

3.2.4.1 Mean time to repair. Applicable.

3.2.4.2 Test points. Not applicable.

3.2.4.3 Shop replaceable assemblies (SRAs). Not applicable.

3.2.4.4 Self test capabilities. Not applicable.

3.2.5 Reliability. The mean-time-between-failures (MTBF) shall be 1000 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Not applicable.

3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection. Applicable. The stand shall be capable of withstanding coolant overpressures of up to 200 psig and nitrogen pressures of up to 150 psig without injury to personnel.

3.2.7.2 Clean air protection. Not applicable.

3.2.7.3 Electrical power protection. Not applicable.

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3.2.8 Calibration cycle. Applicable.

3.3 Performance

3.3.1 General. Applicable. The stand shall provide the capability to support and position liquid-cooled AN/AWG-9 WRAs and AN/AWM-23 panels listed in Table I for draining and refilling of liquid coolant at the intermediate level of maintenance.

3.3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Not applicable. Intermediate level maintenance capability will consist of the following:

a. Mounting and supporting the units listed in Table I for drain, fill, or repair.

b. Connecting the unit to be serviced to the liquid coolant supply or the nitrogen supply.

c. Removing the coolant by means of nitrogen pressure through a hydraulic hose to a storage reservoir.

d. Operating liquid coolant supply or the nitrogen supply controls while retaining the unit in various positions as required.

3.3.4 Warm-up time. Not applicable.

3.4 Detailed functional requirements. Applicable. The stand shall be capable of controlling a coolant flow rate from 0 to 3 gallons per minute at a pressure of 180 pounds per square inch (psi) maximum. The stand pressurization system shall function with nitrogen supplied from the facility at 90 $\pm$5 psi.

3.5 Workmanship. Applicable.

3.6 Elements. Applicable. The stand shall consist of the following functional elements.

3.6.1 Structure. The structure shall enclose and support the other functional elements. The structure shall have a panel to support and position valves, gages, tubing, flex lines, and fittings of the coolant and pressurization systems.

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3.6.2 Coolant hydraulic system. The coolant hydraulic system shall consist of a 200 psi minimum coolant supply valve, two two-way control valves linked together mechanically, flexible supply and return hoses with quick disconnect self-sealing connectors, flow/no-flow sight glass in the return line to permit observation of coolant flow for determining when gas bubbles are present in the coolant, check valve, connecting tubing and fittings, and hose connector adapters for the items under maintenance.

3.6.3 Coolant drain and storage system. The coolant drain and storage system shall consist of one two-way control valve, an 11.7 gallon minimum coolant reservoir, reservoir liquid level sight glass, reservoir 12 psi pressure relief valve plus check valve, reservoir drain valve, sump pan, sump pan drain valve, and connecting tubing and fittings. Capability shall be provided to drain liquid coolant from the reservoir into an external container, or to the liquid cooler reservoir via the facility lines using the facility pressurized nitrogen.

3.6.4 Pressurization system. The pressurization system shall consist of a 200 psi minimum nitrogen supply valve, 100 psi pressure relief valve, 0-200 psi gage, check valve, one two-way control valve, and connecting tubing and fittings.

3.6.5 Work surface. The work surface shall provide the capability to support the units of Table I, except the transmitter and power supplies. (See 3.6.6). The work surface shall have a drain to the sump pan.

3.6.6 Support and positioning mechanism. The support and positioning mechanism shall consist of a rotatable fixture with adapters or clamps, as required, to mount the radar transmitter and power supplies listed in Table I. The mechanism shall have a reversible gear reduction unit which shall not permit movement of the output shaft unless driven from the input end. The mechanism shall be capable of supporting its load in the upright through inverted positions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

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4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table II shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Not applicable. The physical configuration of the stand during acceptance shall be: Fully assembled in accordance with Drawing 493715-1XX.

4.3 Tests

4.3.1 Electrical and electronic items. Not applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The stand is intended to be used at the intermediate maintenance level for servicing the liquid coolant in the AN/AWG-9 WRAs and AN/AWM-23 (V) panels listed in Table I. The stand shall provide the capability to support and position the serviced WRA (or panel) for draining the coolant, for removing entrapped air from the coolant, and for refilling with coolant.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable. In addition the following shall apply.

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TABLE II

ACCEPTANCE TEST PLAN FOR COOLANT SYSTEM FILL-DRAIN STAND

MT-4730 ( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate the functional operation of the coolant system fill-drain stand.

2.0 TEST CONDITIONS

a. Test Equipment

(1) Leak detection fluid.

(2) Pipe and fitting to connect to nitrogen supply.

b. Utilities Required

(1) Nitrogen as specified in Paragraph 3.2.2.6 of this specification.

3.0 TESTS In accordance with Acceptance Requirements Document AR31082-531

a. Operational run-in tests are not required.

b. Perform tests in accordance with the acceptance requirements document.

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6.3.15 Nonstandard. Parts, materials and processes are defined as nonstandard if they are not covered by the documents listed in ANA BULLETIN 400, Airborne Electronic and Associated Equipment, Applicable Documents.

6.4 Nomenclature cross-reference. The following is a cross-reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

	Official Nomenclature	Contractor Part Number
a.	Oscillator, Radio Frequency O-1633( )/AWG-9	481001-1XX
b.	Transmitter, Radar T-1224( )/AWG-9	481011-1XX
c.	Power Supply PP-6770 ( )/AWG-9	481013-1XX
d.	Power Supply PP-6771 ( )/AWG-9	481014-1XX
e.	Receiver, Radar R-1785 ( )/AWG-9	481022-1XX
f.	Receiver, Infrared R-1786 ( )/AWG-9	481101-1XX
g.	Processor, Data CV-2947 ( )/AWG-9	481120-1XX
h.	Cooler, Liquid, Electronic Equipment, HD-908 ( )/AWM-23(V)	493701-1XX
i.	Cooler, Liquid, Electronic Equipment HD-957 ( )/AWM-23(V)	493711-1XX
j.	Stand, Fill-Drain, Coolant System MT-4730 ( )/AWM-23(V)	493715-1XX
k.	Dummy Load, Electrical DA-630 ( )/AWM-23(V)	3255628-XXX
l.	Dummy Load, Electrical DA-631 ( )/AWM-23(V)	3069350-XXX
m.	Dummy Load, Electrical DA-632() DA-632 ( )/AWM-23(V)	3069320-XXX
n.	Transducer TR-283 ( )/AWM-23(V)	3070775

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6.5 Liquid cooler. Applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of this specification are the same as MIL-T-81734 (AS) the word "applicable" is used in this specification. When the term "not applicable" is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met. See 6.6 of MIL-T-81734(AS).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply:

PSI	Pounds per square inch

6.8 Convenience outlet power. Not applicable.

Project No. 4920-N338

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-81996(AS) STAND, FILL-DRAIN, COOLANT SYSTEM MT-4730( )/AW,-23(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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